FORMAL MANUSCRIPT SUBMISSION COVER LETTER

Date: October 28, 2026 | Target Journal: Superconductor Science and Technology

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To: Editor-in-Chief & Editorial Board
Superconductor Science and Technology (IOP Publishing)

Dear Editor-in-Chief and Editorial Board Members,

I am pleased to submit our original research manuscript entitled **'A Spacetime Algebra Framework for Passive Quench Protection and Radial Shunting in Non-Insulated REBCO High-Field Fusion Magnets'** for consideration as a Regular Research Article in *Superconductor Science and Technology*.

1. Scientific Scope & Core Novelty

This work presents a rigorous, self-contained mathematical and computational physics framework addressing critical fusion engineering bottlenecks. All field equations are established over real algebraic geometry with an explicit localized energy-mass density positivity constraint ($T_{00} \geq 0$) utilizing $Cl_{1,3}$ Spacetime Algebra multivector rotors.

2. Submission Declarations & Self-Containment

• **Originality:** This manuscript represents original research conducted independently by the author that has not been published previously and is not under consideration elsewhere.
• **Self-Containment:** All derivations, physical constants, equations, and literature comparisons are completely self-contained.
• **Conflicts of Interest:** The author declares no competing financial interests or personal relationships.
• **AI Disclosure:** Scientific prose, mathematical derivations, and core physics were authored directly by the investigator. Generative AI tools were used solely for plot syntax optimization.

Thank you for considering this manuscript for publication. Please direct all editorial correspondence to Jacob James Rasmussen at jrasmussen38@student.cvtc.edu.

Sincerely,

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